

KRASNOZERSKOE, Russian Federation

WMO: 298130

Lat: **53.989N**Lon: **79.253E**Elev: **144**StdP: **99.61**Time Zone: **7.00 (E07)**Period: **94-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.5	-32.0	-39.0	0.1	-34.3	-35.4	0.1	-31.6	11.6	-7.4	10.3	-9.2	1.9	210	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	30.4	18.5	28.6	18.5	26.9	17.8	20.8	26.6	19.9	25.6	19.0	24.8	4.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.0	23.0	17.9	13.1	22.4	16.9	12.3	21.7	61.1	26.5	57.8	25.5	54.9	24.8	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.1	8.0	DB	-36.8	33.4	3.6	1.9	-39.4	34.8	-41.5	35.9	-43.5	37.0	-46.0	38.4	(4)
(5)			WB	-37.3	22.6	3.8	1.4	-40.0	23.6	-42.2	24.4	-44.4	25.2	-47.1	26.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-18.1	-15.5	-6.7	4.6	12.5	18.6	19.6	17.1	11.3	3.4	-6.2	-14.0	(6)	
	DBStd	14.59	8.24	7.02	6.10	6.12	5.37	4.07	3.26	3.77	4.94	5.20	7.12	8.01	(7)	
	HDD10.0	3808	870	714	519	182	37	1	0	1	42	211	487	744	(8)	
	HDD18.3	6014	1128	948	777	412	191	46	24	68	218	462	737	1003	(9)	
	CDD10.0	1003	0	0	0	19	115	259	299	222	80	8	0	0	(10)	
	CDD18.3	167	0	0	0	0	11	54	64	31	7	0	0	0	(11)	
	CDH23.3	1571	0	0	0	7	151	485	516	323	89	1	0	0	(12)	
	CDH26.7	419	0	0	0	1	35	133	139	83	28	0	0	0	(13)	
(14) Wind	WSAvg	3.6	3.6	3.8	4.1	3.9	4.0	3.3	2.9	3.0	3.3	3.4	3.9	4.1	(14)	
(15) Precipitation	PrecAvg	348	17	13	14	17	26	44	66	42	30	30	26	21	(15)	
	PrecMax	464	35	27	30	39	55	102	160	125	69	78	62	47	(16)	
	PrecMin	216	3	0	2	1	0	10	15	3	2	7	13	6	(17)	
	PrecStd	76	9	7	8	10	13	22	38	28	18	16	11	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	0.5	6.2	24.0	30.1	31.9	32.6	31.9	30.4	20.0	8.7	1.6	(19)	
		MCWB	0.3	-0.4	2.7	12.7	16.4	19.0	19.9	18.6	16.5	11.7	6.7	0.4	(20)	
	2%	DB	-1.8	-1.1	4.0	20.2	27.1	30.0	30.0	28.9	25.6	16.5	5.9	0.2	(21)	
		MCWB	-2.4	-1.8	1.7	11.3	15.4	18.2	19.5	18.7	14.9	9.9	3.9	-0.4	(22)	
	5%	DB	-3.9	-2.9	2.6	17.3	24.8	28.1	28.0	26.7	22.4	13.5	3.9	-1.5	(23)	
		MCWB	-4.6	-3.6	0.8	10.0	14.5	18.2	19.4	17.8	13.8	8.5	2.5	-2.2	(24)	
	10%	DB	-5.9	-5.1	1.2	14.2	22.0	26.3	26.2	24.8	19.6	10.8	1.7	-3.4	(25)	
		MCWB	-6.5	-5.7	-0.1	8.3	13.1	17.6	18.9	16.7	12.6	7.0	0.5	-4.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	-0.3	3.4	14.1	18.4	21.6	22.2	21.8	17.7	12.5	7.1	0.6	(27)	
		MCDB	0.7	0.2	5.4	21.5	26.7	29.2	27.3	27.0	26.9	17.9	8.2	1.0	(28)	
	2%	WB	-2.3	-1.8	2.0	12.3	16.8	20.0	21.1	19.8	15.8	10.7	4.2	-0.3	(29)	
		MCDB	-1.8	-1.2	3.7	19.0	25.2	27.0	26.6	25.7	23.6	15.7	5.6	0.2	(30)	
	5%	WB	-4.5	-3.6	1.0	10.4	15.2	19.1	20.3	18.7	14.4	9.0	2.4	-2.2	(31)	
		MCDB	-4.0	-2.9	2.4	16.4	22.7	25.8	25.6	25.0	21.2	12.4	3.8	-1.6	(32)	
	10%	WB	-6.5	-5.6	0.0	8.7	13.9	18.2	19.5	17.6	13.1	7.4	0.7	-4.0	(33)	
		MCDB	-5.9	-5.1	1.2	13.7	21.2	24.4	24.6	23.5	18.6	10.7	1.7	-3.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.2	8.0	9.7	11.4	10.4	10.0	11.0	10.3	7.8	6.3	7.4	(35)	
		MCDBR	8.1	7.8	7.5	14.1	14.7	13.2	12.4	14.0	15.1	12.2	6.8	7.7	(36)	
	5% WB	MCWBR	8.1	7.7	6.2	7.8	6.7	5.4	4.8	5.6	6.7	6.7	5.4	7.7	(37)	
		MCWBR	8.0	7.6	6.7	13.2	12.9	11.2	10.3	12.1	13.1	11.0	6.7	7.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.294	0.326	0.380	0.405	0.407	0.410	0.436	0.418	0.371	0.352	0.328	0.297	(40)	
		taud	2.021	1.972	1.913	2.138	2.201	2.278	2.200	2.222	2.304	2.236	2.082	1.977	(41)	
	Ebn at Noon		617	726	768	817	838	840	807	797	791	705	567	511	(42)	
		Edh at Noon	74	110	147	135	133	125	133	123	99	85	70	64	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.04	2.12	3.60	4.72	5.75	6.14	5.56	4.72	3.33	1.74	0.83	0.69	(44)	
	RadStd		0.09	0.11	0.28	0.32	0.49	0.47	0.38	0.36	0.32	0.21	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-0.59	N/A	N/A	N/A	N/A	(46)
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page